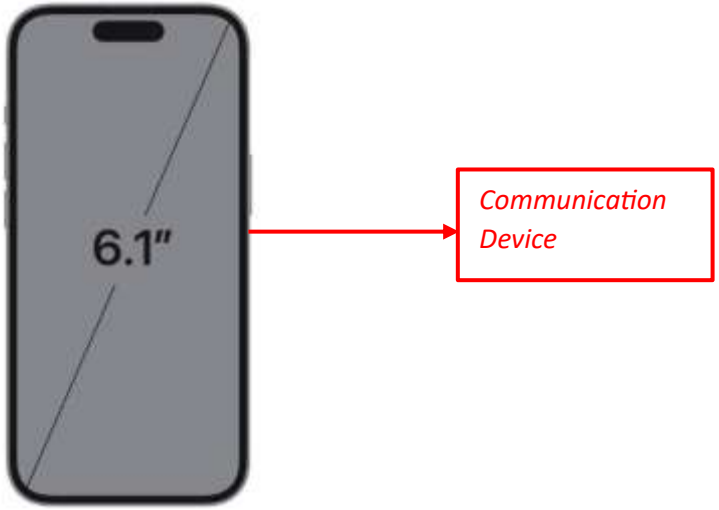
Infringement Chart Strength: Medium to High

Priority Date: 05, April 2006

Product name: FordPass Connect

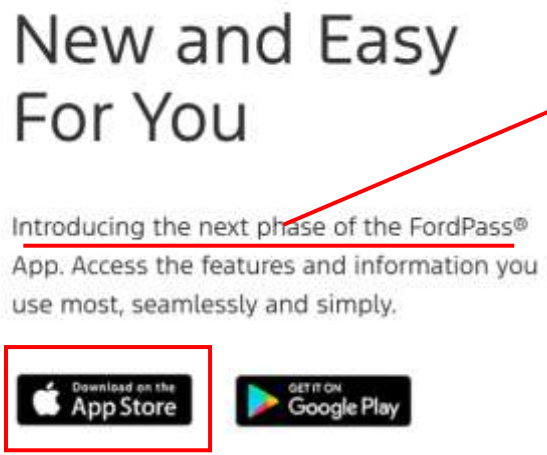
Product Release date: 2016

Note: Few logical inferences are taken to create the chart. The inferences are related to the "Receiver receives the signal", "receiver of the communication device", and "transceiver of product"

US9096189B2	FordPass Connect
1. A communication device of at least one of a cell phone, a smart phone, a desktop, a handheld, a PDA, a laptop, or a computer terminal for monitoring products, interconnected to a product for communication therebetween, comprising:	*FordPass app works on the communication device of at least one smartphone (i.e., iPhone), for monitoring products (i.e., vehicles), interconnected to a product for communication* *The FordPass app is compatible with iPhone devices, enabling vehicle monitoring and remote-control functions from any location. FordPass provides notifications or alerts about the vehicle's information and door lock and unlock status.*  Source: https://www.apple.com/iphone-15/specs/

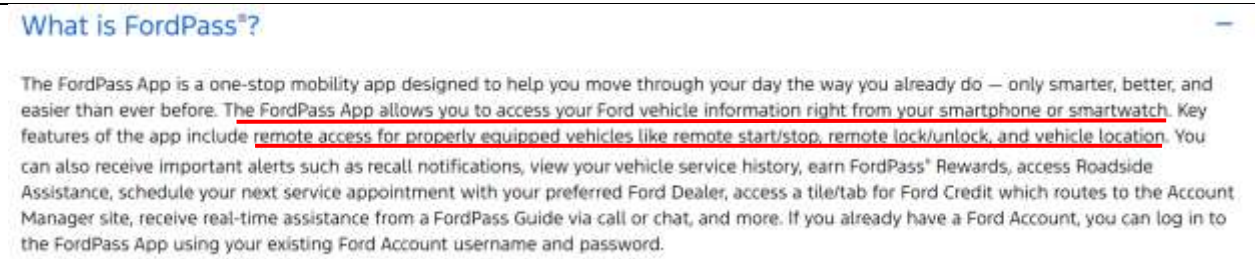
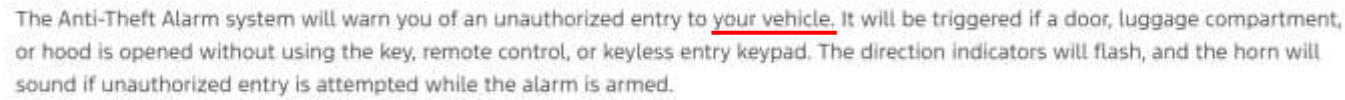


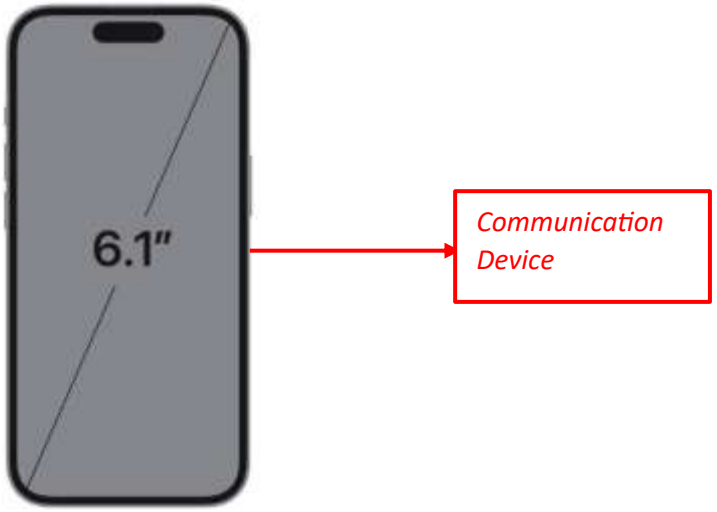
Source: https://apps.apple.com/us/app/fordpass/id1095418609



FordPass Download app store for communication device

Source: https://www.ford.com/support/category/fordpass/

	 Source: Link  Source: https://www.ford.com/support/how-tos/keys-and-locks/door-locks-and-alarms/how-do-i-use-the-anti-theft-alarm-system/
at least one of a central processing unit (CPU) for executing and carrying out the instructions of a computer program, a network processor which is specifically targeted at the networking application domain, or a front end processor for communication between a host computer and other devices;	*The FordPass app works on iPhone devices, an iPhone with a central processing unit (CPU) for executing and carrying out the instructions (e.g., remote lock or unlock) of a computer program.* *The FordPass app is compatible with iPhone devices, leveraging the device's central processing unit (CPU) to execute tasks and process commands. Through this, the app is able to send signals to the vehicle, enabling remote control functions such as door locking and unlocking.*





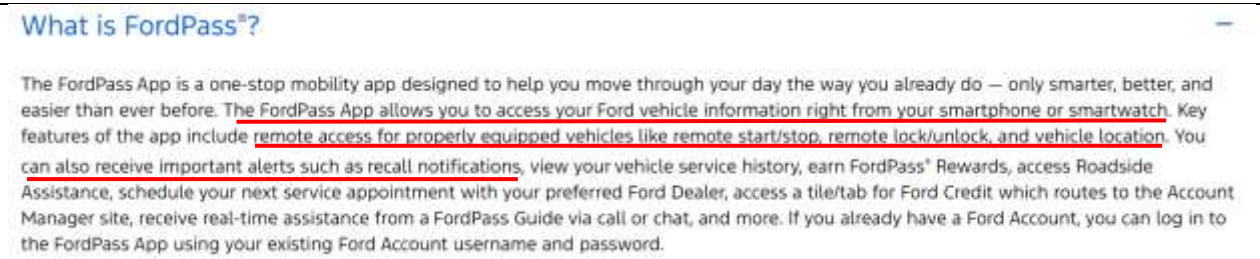
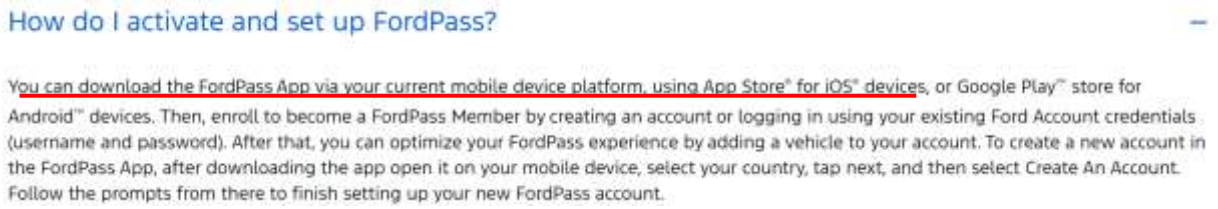
A16 Bionic chip

6-core CPU with 2 performance and 4 efficiency cores

5-core GPU

16-core Neural Engine

Source: https://www.apple.com/iphone-15/specs/

	 Source: Link
a transmitter for transmitting signals and messages to at least one of plurality product groups based on the categories of a multi-sensor detection device, a maritime cargo container, a cell phone detection device, or a locking device;	*The communication device transmitter (e.g., Bluetooth) for transmitting signals and messages (e.g., Door Lock or unlock) to the vehicle at least one of the multi-sensor detection devices.* *The FordPass app is compatible with the iPhone and can function as a transmitter, using Bluetooth technology, to send commands or signals to a vehicle. This allows the user to perform actions such as locking or unlocking the vehicle's doors. The multi-sensor detection devices in the vehicle are responsible for recognizing these signals and executing the corresponding actions.*  Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/



Communication Device

5G (sub-6 GHz and mmWave) with 4x4 MIMC

Gigabit LTE with 4x4 MIMO and LAA[10]

Wi-Fi 6 (802.11ax) with 2x2 MIMO

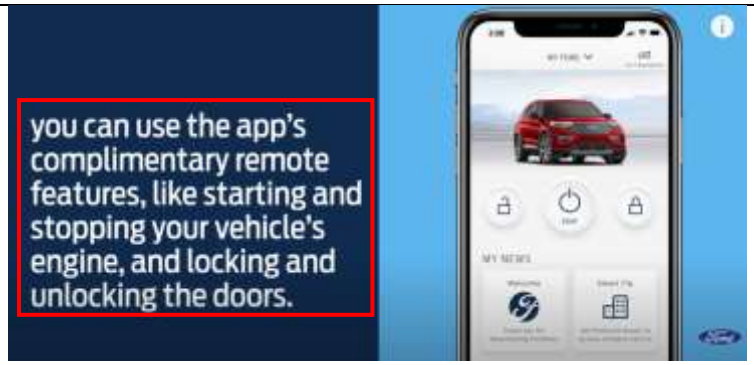
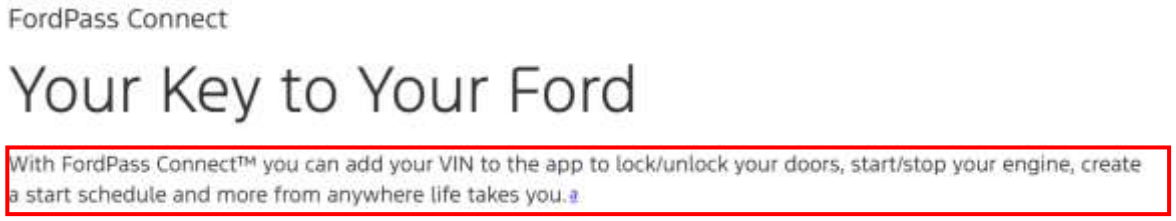
Bluetooth 5.3 → *transmitter*

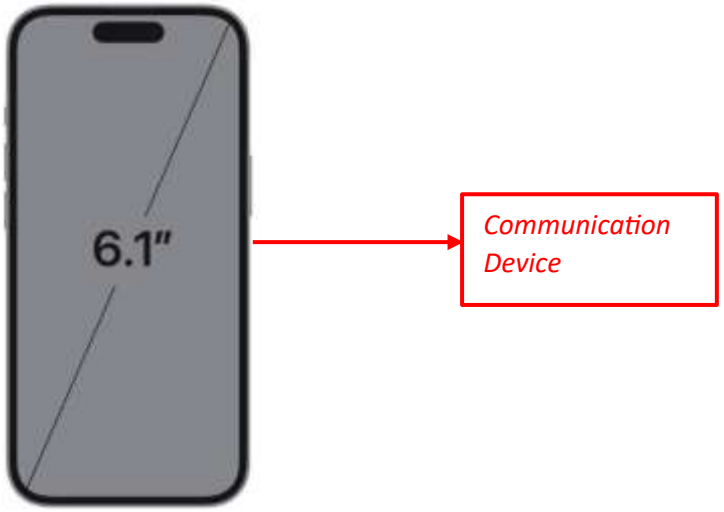
Second-generation Ultra Wideband chip[11]

NFC with reader mode

Express Cards with power reserve

Source: https://www.apple.com/iphone-15/specs/

	 Source: https://youtu.be/lIvw07oG6gQ?t=22  Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/
a receiver for receiving signals, data or messages from at least one of plurality product groups based on the categories of a multi-sensor detection device, a maritime cargo container, a cell phone detection device, or a locking device;	*The communication device is a receiver for receiving signals, data, or messages (i.e., warning and notification) from at least one of the multi-sensor detection devices.* *The FordPass app is compatible with iPhone devices and can function as a receiver to receive notifications and warning messages from vehicles. Users receive important vehicle alerts directly on their smartphones. The multi-sensor detection devices in the vehicle are responsible for recognizing these signals and executing the corresponding actions. Therefore we can infer that the receiver receiving a signal.*



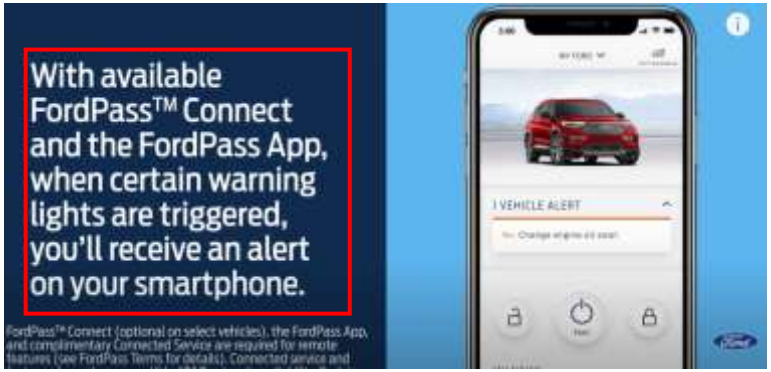
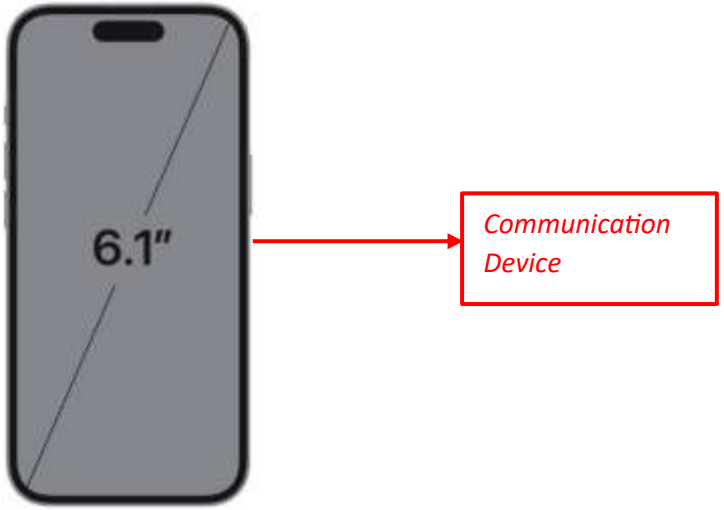
Source: https://www.apple.com/iphone-15/specs/

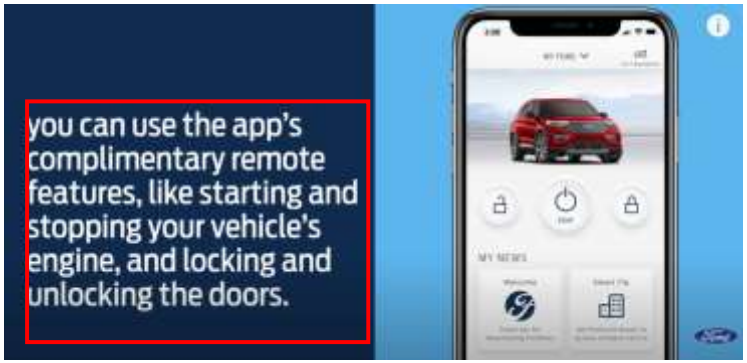
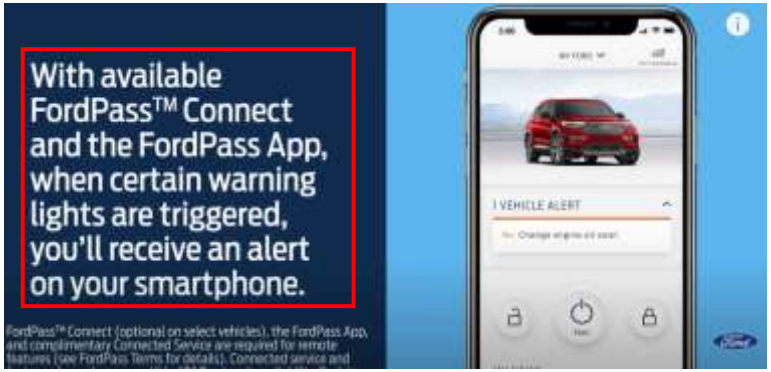
How do I activate and set up FordPass?

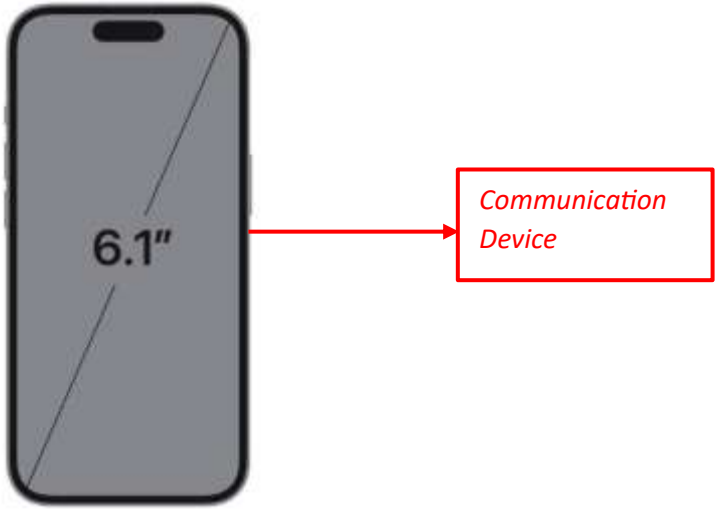
You can download the FordPass App via your current mobile device platform, using App Store® for iOS® devices, or Google Play™ store for Android™ devices. Then, enroll to become a FordPass Member by creating an account or logging in using your existing Ford Account credentials (username and password). After that, you can optimize your FordPass experience by adding a vehicle to your account. To create a new account in the FordPass App, after downloading the app open it on your mobile device, select your country, tap next, and then select Create An Account. Follow the prompts from there to finish setting up your new FordPass account.

The FordPass App is a one-stop mobility app designed to help you move through your day the way you already do — only smarter, better, and easier than ever before. The FordPass App allows you to access your Ford vehicle information right from your smartphone or smartwatch. Key features of the app include remote access for properly equipped vehicles like remote start/stop, remote lock/unlock, and vehicle location. You can also receive important alerts such as recall notifications, view your vehicle service history, earn FordPass® Rewards, access Roadside Assistance, schedule your next service appointment with your preferred Ford Dealer, access a tile/tab for Ford Credit which routes to the Account Manager site, receive real-time assistance from a FordPass Guide via call or chat, and more. If you already have a Ford Account, you can log in to the FordPass App using your existing Ford Account username and password.

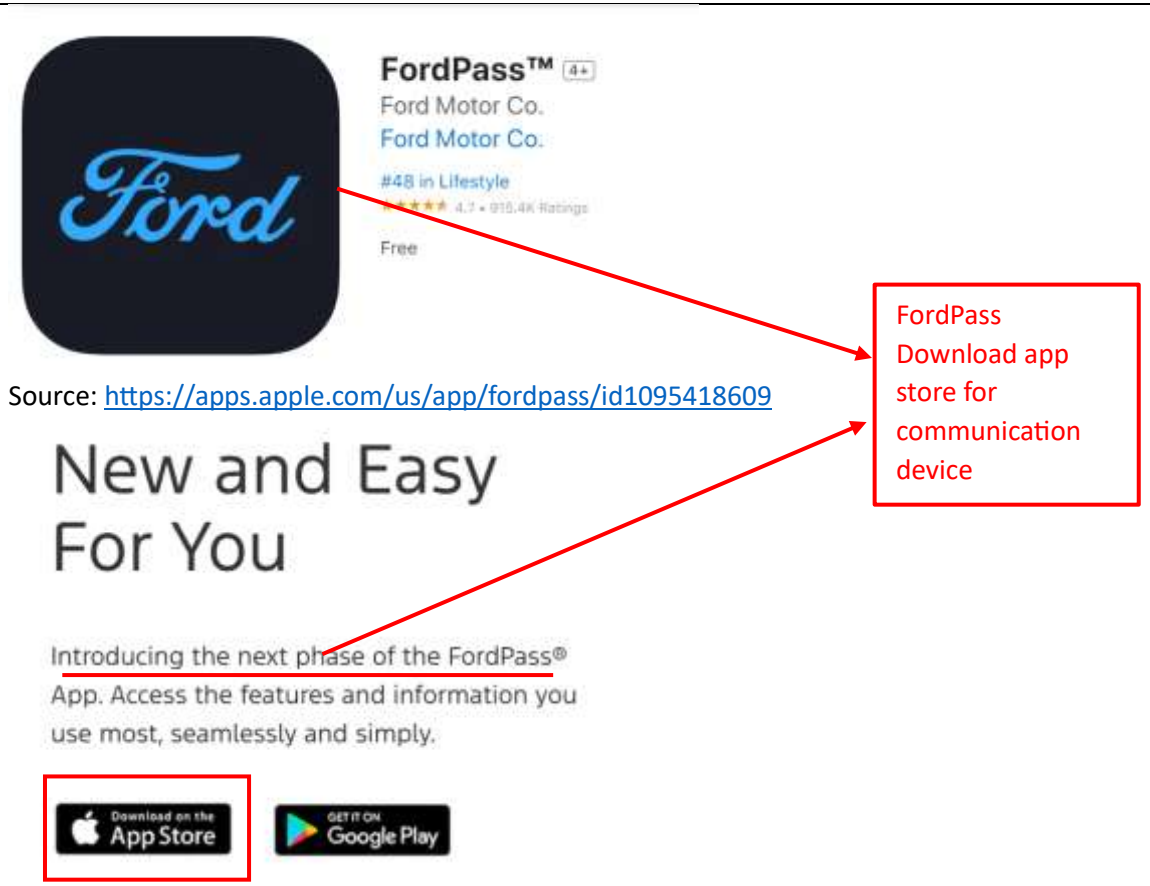
Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/

	 Source: https://youtu.be/ZK32KyDBZIY?t=23
at least one satellite connection, Bluetooth connection, WiFi connection, internet connection, radio frequency (RF) connection, cellular connection, broadband connection, long and short range radio frequency (RF) connection, or GPS connection;	*The iPhone is equipped with at least one cellular connection, allowing it to transmit and receive signals for data exchange and remote communication purposes.* 

	Source: https://www.apple.com/iphone-15/specs/ "FordPass" Connect, the FordPass" App., and complimentary Connected Service are required for remote features (see FordPass Terms for details). Connected service and features depend on compatible AT&T network availability. Evolving technology/cellular networks/vehicle capability may limit functionality and prevent operation of connected features. Connected service excludes Wi-Fi" hotspot Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/   Source: https://youtu.be/IIvw07oG6gQ?t=22

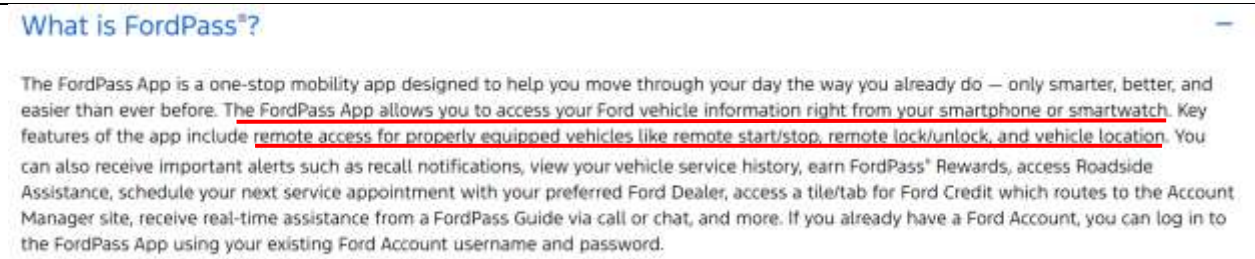
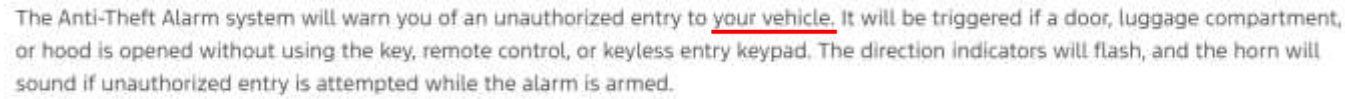
the communication device is at least a fixed, portable or mobile communication device interconnected to a fixed, portable or mobile product, capable of wired or wireless communication therebetween; and	*The communication device is at least a portable or mobile communication device (e.g., smartphone) interconnected to a portable (e.g., vehicle) capable of wireless communication (e.g., Bluetooth).* *The FordPass application is designed for compatibility with iPhone devices, functioning as a communication interface to transmit and receive signals to and from portable vehicles. This allows for remote vehicle management and control features, leveraging wireless communication protocols to facilitate interaction between the mobile application and the vehicles.* 

	5G (sub-6 GHz and mmWave) with 4x4 MIMC Gigabit LTE with 4x4 MIMO and LAA[10] Wi-Fi 6 (802.11ax) with 2x2 MIMO Bluetooth 5.3 → *Wireless Communications* Second-generation Ultra Wideband chip[11] NFC with reader mode Express Cards with power reserve Source: https://www.apple.com/iphone-15/specs/



Source: https://apps.apple.com/us/app/fordpass/id1095418609

Source: https://www.ford.com/support/category/fordpass/

	 Source: Link  Source: https://www.ford.com/support/how-tos/keys-and-locks/door-locks-and-alarms/how-do-i-use-the-anti-theft-alarm-system/
whereupon the communication device, is interconnected to a product equipped to receive signals from or send signals to lock or unlock doors, activate or deactivate security systems, activate or deactivate multi-sensor detection systems, or to activate or deactivate cell phone detection systems;	*The communication device (e.g., iPhone) is interconnected to a product (e.g., vehicle) equipped to receive signals from or send signals to lock or unlock doors, activate or deactivate security systems, activate or deactivate multi-sensor detection systems (e.g., Disarming the alarm)* *The FordPass app is compatible with iPhone devices and is interconnected with vehicles. The FordPass app facilitates the transmission and reception of signals, such as commands for locking and unlocking doors and disabling the vehicle alarm*



Communication Device

Source: https://www.apple.com/iphone-15/specs/

What is FordPass®?

The FordPass App is a one-stop mobility app designed to help you move through your day the way you already do — only smarter, better, and easier than ever before. The FordPass App allows you to access your Ford vehicle information right from your smartphone or smartwatch. Key features of the app include remote access for properly equipped vehicles like remote start/stop, remote lock/unlock, and vehicle location. You can also receive important alerts such as recall notifications, view your vehicle service history, earn FordPass® Rewards, access Roadside Assistance, schedule your next service appointment with your preferred Ford Dealer, access a tile/tab for Ford Credit which routes to the Account Manager site, receive real-time assistance from a FordPass Guide via call or chat, and more. If you already have a Ford Account, you can log in to the FordPass App using your existing Ford Account username and password.

Source: Link

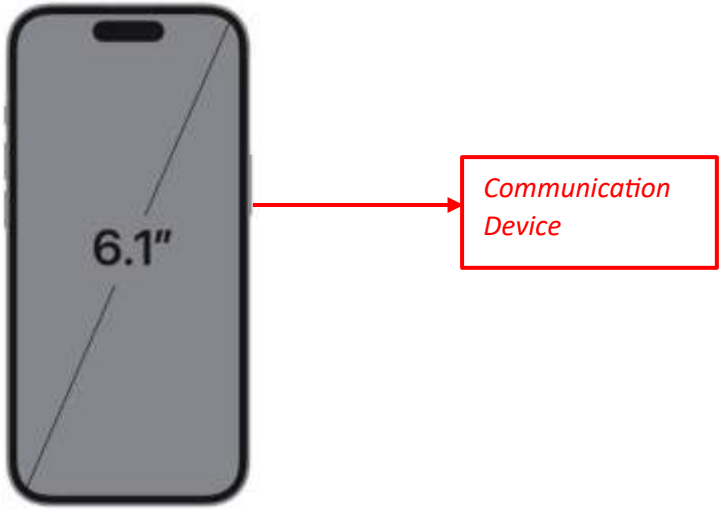
Remote lock or lock

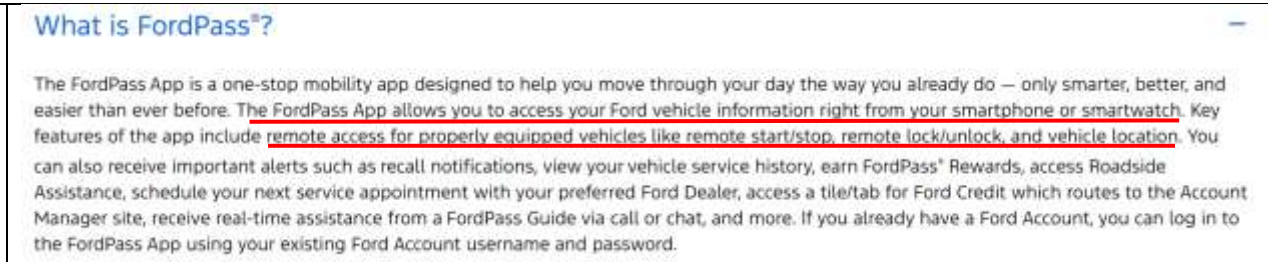
Disarming the Alarm

Disarm the alarm by any of the following actions:

- Unlock the doors or luggage compartment with the remote control or keyless entry keypad.
- Switch the ignition on or start the vehicle.
- Use a key in the driver's door to unlock the vehicle, then switch the ignition on within 12 seconds.

security systems active or Deactivate

	Source: https://www.ford.com/support/how-tos/keys-and-locks/door-locks-and-alarms/how-do-i-use-the-anti-theft-alarm-system/
wherein the communication device receives a signal via any of one or more products listed in any of the plurality of product grouping categories;	*The FordPass app works on the communication device receiving (e.g., iPhone or smartphone) a signal via any of one or more products (i.e., Vehicle) listed in any of the plurality of product grouping categories.* *The FordPass app is compatible with iPhone devices and enables the transmission and reception of alert information and notifications to and from the vehicle.*  Source: https://www.apple.com/iphone-15/specs/

	 Source: Link
wherein at least one satellite connection, Bluetooth connection, WiFi connection, internet connection, radio frequency (RF) connection, cellular connection, broadband connection, long and short range radio frequency (RF) connection is capable of signal communication with the transmitter and the receiver of the communication device and transceivers of the products;	*The communication device used cellular connection (e.g., AT&T technology) connection is capable of signal communication with the transmitter (e.g., Bluetooth) and the receiver of the communication device and transceivers of the products (e.g., vehicle)* *The sole permissible communication type between Bluetooth (e.g., Transmitter) and the receiver of the communication device, along with the transceivers of the vehicle is integrated into the vehicle and encompasses functions such as alert notifications and door locking/unlocking. This communication is limited to methods selected from a group that includes cellular technologies, specifically utilizing AT&T's network infrastructure. Therefore, we can infer that the receiver of the communication and transmitter of the Product.*



Communication Device

5G (sub-6 GHz and mmWave) with 4x4 MIMC

Gigabit LTE with 4x4 MIMO and LAA[10]

Wi-Fi 6 (802.11ax) with 2x2 MIMO

Bluetooth 5.3 → *transmitter*

Second-generation Ultra Wideband chip[11]

NFC with reader mode

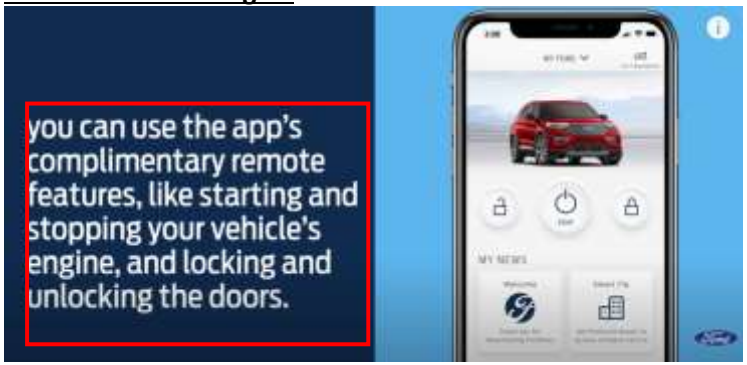
Express Cards with power reserve

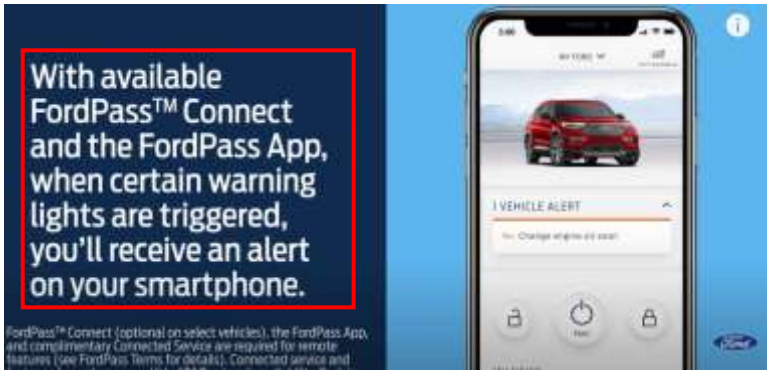
Source: https://www.apple.com/iphone-15/specs/

"FordPass" Connect, the FordPass" App., and complimentary Connected Service are required for remote features (see FordPass Terms for details). Connected service and features depend on compatible AT&T network availability. Evolving technology/cellular networks/vehicle capability may limit functionality and prevent operation of connected features. Connected service excludes Wi-Fi" hotspot

Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/

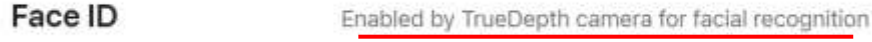
Send and Receive signal





Source: https://youtu.be/lIvw07oG6gQ?t=22

wherein the communication device is equipped with a biometric lock disabler that incorporates at least one of a fingerprint recognition, voice recognition, face recognition, hand geometry, retina scan, iris scan and signature such that the communication device that is at least one of the cell phone, the smart phone, the desktop, the handheld, the PDA, the laptop or the computer terminal is locked by the biometric lock disabler to prevent unauthorized use;	*The communication device (e.g., iPhone) is equipped with a biometric lock disabler that incorporates at least one voice recognition, and face recognition such that the smartphone is locked by the biometric lock disabler to prevent unauthorized use.* *The FordPass app is compatible with iPhone devices and integrates biometric security features, including voice recognition and facial recognition. These features protect the device against unauthorized access attempts to unlock the vehicle by sending a signal, utilizing a biometric lock disabler to prevent unauthorized.*  *Communication Device* Siri[14] Get help with everyday tasks like sending messages, setting reminders, and more Activate hands-free with only your voice using "Siri" or "Hey Siri" Protected by the strongest privacy of any intelligent assistant Learn more about Siri ›

	 Source: https://www.apple.com/iphone-15/specs/ How do I activate and set up FordPass? – You can download the FordPass App via your current mobile device platform, using App Store℠ for iOS℠ devices, or Google Play™ store for Android™ devices. Then, enroll to become a FordPass Member by creating an account or logging in using your existing Ford Account credentials (username and password). After that, you can optimize your FordPass experience by adding a vehicle to your account. To create a new account in the FordPass App, after downloading the app open it on your mobile device, select your country, tap next, and then select Create An Account. Follow the prompts from there to finish setting up your new FordPass account. The FordPass App is a one-stop mobility app designed to help you move through your day the way you already do — only smarter, better, and easier than ever before. The FordPass App allows you to access your Ford vehicle information right from your smartphone or smartwatch. Key features of the app include remote access for properly equipped vehicles like remote start/stop, remote lock/unlock, and vehicle location. You can also receive important alerts such as recall notifications, view your vehicle service history, earn FordPass℠ Rewards, access Roadside Assistance, schedule your next service appointment with your preferred Ford Dealer, access a tile/tab for Ford Credit which routes to the Account Manager site, receive real-time assistance from a FordPass Guide via call or chat, and more. If you already have a Ford Account, you can log in to the FordPass App using your existing Ford Account username and password. Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/
wherein the only type or types of communication with the transmitter and the receiver of the communication device and transceivers of the products is a type or types selected from the group consisting of satellite, Bluetooth, WiFi, internet, radio frequency (RF), cellular, broadband, and long	*The one only type or types of communication (e.g., alert notification and lock unlock the door) with the transmitter (e.g., Bluetooth) and the receiver of the communication device and transceivers of the products is a type or types selected from the group consisting of cellular (e.g., AT&T technology).* *The sole permissible communication type between Bluetooth (e.g., Transmitter) and the receiver of the communication device, along with the transceivers of the vehicle is integrated into the vehicle and encompasses functions such as alert notifications and door locking/unlocking. This communication is limited to methods selected from a group that includes cellular technologies, specifically utilizing AT&T's network infrastructure. Therefore, we can infer that the receiver of the communication and transmitter of the Product.*

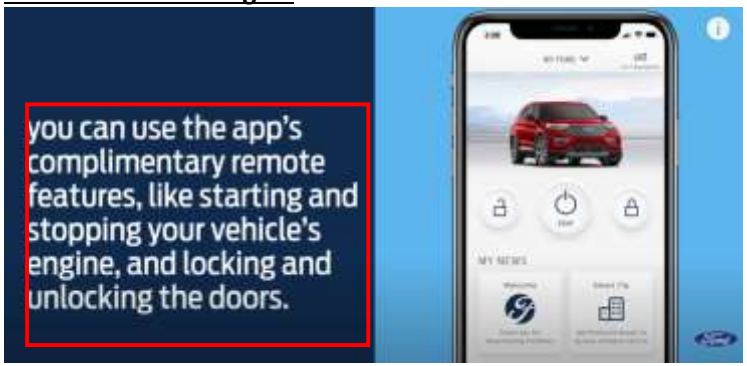
and short range radio frequency (RF).	 *Communication Device* 5G (sub-6 GHz and mmWave) with 4x4 MIMC Gigabit LTE with 4x4 MIMO and LAA[10] Wi-Fi 6 (802.11ax) with 2x2 MIMO Bluetooth 5.3 → *transmitter* Second-generation Ultra Wideband chip[11] NFC with reader mode Express Cards with power reserve Source: https://www.apple.com/iphone-15/specs/

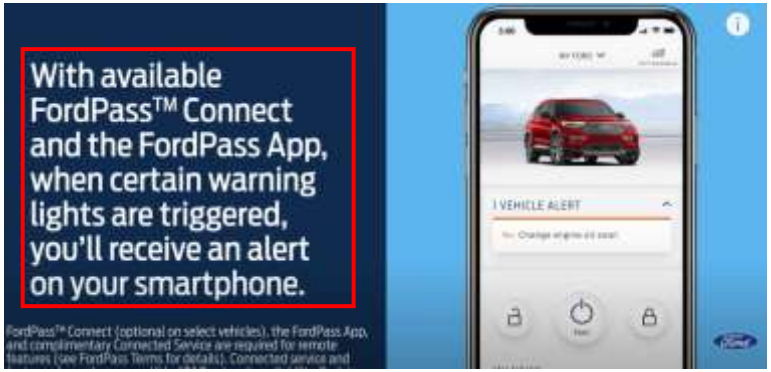
Cellular network

"FordPass" Connect, the FordPass" App., and complimentary Connected Service are required for remote features (see FordPass Terms for details). Connected service and features depend on compatible AT&T network availability. Evolving technology/cellular networks/vehicle capability may limit functionality and prevent operation of connected features. Connected service excludes Wi-Fi" hotspot

Source: https://www.ford.com/support/category/fordpass/fordpass-sync-connect/

Send and Receive signal





Source: https://youtu.be/IIvw07oG6gQ?t=22

Bayslope